

PROFESSIONAL SUMMARY

Recent B.Tech Computer Science graduate with strong fundamentals in data structures, algorithms, and software engineering, demonstrated through building a custom language interpreter, algorithmic game logic, and deployed backend systems. Proficient in C, Python, FastAPI, and React.js.

EDUCATION

Graphic Era Hill University, Dehradun
B.Tech in Computer Science and Engineering

Aug 2022 – May 2026
CGPA: 7.22

TECHNICAL SKILLS

Languages: C, C++, Java, Python, SQL

Frontend: React.js, Vite

Backend: Flask, FastAPI, Supabase (Auth, Database)

Databases: PostgreSQL, SQLite

Tools: Git, GitHub, VS Code, Vercel, Render

PROJECTS

Picoc Interpreter | C, Flask, CodeMirror, Prism.js

[GitHub](#) | [Live Demo](#)

- Built a custom lexer, recursive-descent parser, and AST evaluator from scratch in C.
- Supported multi-language syntax parsing across C, C++, Java, and Python subsets.
- Implemented line-level error tracking to pinpoint syntax errors during evaluation.
- Built a Flask backend connecting the C engine to a CodeMirror-based web UI.

Ultimate Tic-Tac-Toe | React.js, Vite, Supabase Auth, PostgreSQL

[GitHub](#) | [Live Demo](#)

- Designed an algorithm to procedurally generate win conditions for any grid size.
- Built a scaled MMR scoring system with win/loss points adjusted by grid difficulty.
- Implemented dynamic 3x3/4x4/5x5 grids and a live Blitz Mode countdown timer.
- Built a Supabase-backed system for authentication and persistent player profiles.

Air Quality Predictor | React.js, FastAPI, Random Forest, SQLite

[GitHub](#) | [Live Demo](#)

- Built a Random Forest sliding-window model to forecast PM2.5 with an R^2 of 0.89.
- Integrated Open-Meteo APIs with FastAPI and SQLite for real-time air quality data.
- Developed a React dashboard with Leaflet maps and Chart.js visualizations.
- Deployed the application on Render, enabling live access to real-time PM2.5 forecasts.

CERTIFICATIONS

- Completed the Cybersecurity Job Simulation Analyst certification.
- Earned the Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate certification.
- Completed Introduction to Generative AI by Google Cloud & Simplilearn SkillUp.

COURSEWORK SKILLS

- Data Structures and Algorithms (DSA)
- Machine Learning (ML)
- Database Management Systems (DBMS)
- Object-Oriented Programming (OOP)
- Computer Networks (CN)
- Software Development Life Cycle (SDLC)
- Operating Systems (OS)